Specification

VFxxxxxxxxxxx

Specification Book

Supersedes Edition (Doc.No) (mm.yyyy)

Scope

Applicable Document of Part 1
Common Electrical Specification for Supplier

Security Access Algorithm

Range Of Validity

VinFast

Author:

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Approval Confirmed By

Function	Name	Signature	Date

Change Order KAU-Number

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1 General Information

1.1 Language

This Specification is made only in English language version.

1.2 Change History

Version	Chapter	Brief Description	Date	Edited By
1.0.0	All	Initial Version	2019.10.14	Nguyen Hoang Long
1.0.1	All	PDF to word document	*25.12.2019*	Rasul Nadaf
1.0.2	3.5	Update diagram for failure behavior and added notes for diagram to have better readability	*21.08.2020*	Rasul Nadaf
1.03	3.5	Corrected mistakes in the diagram	*20.05.2022*	Ashad Qureshi

Table 1: Change History

1.3 Versioning

To determine the actual status of the present specification, see the following table:

Version	State	Gate/Milestone
0.1.0 - 0.49.9	DRA „Draft“	FUP
0.50.0 - 0.99.9	ICS „Inquiry Concept Specification“	ILF
1.0.0 - 1.99.9	ISS „Inquiry Serial Specification“	AF
3.0.0	FIN „Final“	TA

Table 2: Rules for Versioning

Version Digit description:

- **First Digit**: Major Release (Milestone Release)
- **Second Digit**: Minor Release (Function description change)
- **Third Digit**: No functional content change (e.g. spelling correction)

2 Introduction

2.1 Purpose of the Document

This document describes the basic principle of operation for the UDS Service $27 Security Access and the using in the project. Any ECUs and the Test system which use the Service Security Access, shall fulfil the requirements of this document.

Note:
For all ECUs, which are not able to fulfil the requirements of this Specification, a deviation must be request and agreed by the responsible system design department. All deviations must be documented. Only then the deviations can approve.

2.2 Terms and definitions

The following list of terms and definitions are used throughout this document with the given meaning.

2.3 Abbreviations

Abbreviation	E/E Component Name
ECU	Electronic Control Unit
DID	Data Identifier
PID	Parameter Identifier
MAC	Message Authentication Code
HMAC	Hash-based Message Authentication Code
CRC	Cyclic Redundancy Check
M	Mandatory
O	Optional
U	User Conditional
RND	Random Number
NRC	Negative Response Code

Table 3: List of abbreviations

2.4 Definitions

2.4.1.1 Obligation of Requirement Wording

The following wording rules are used throughout this document.

Shall

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

Should

The word "should" is used for preferences or desires in implementation that are not strictly binding.

May

The word "may" is used to denote options that are left to the implementer.

3 System Description

3.1 Overview Basic Services and Subfunctions

Basically, the Services Security Access is an optional service. It is used in control devices, which need a particular security against access by unauthorized users and Tester equipment.

The implementation of this Service has to follow the requirements of the ISO 14229-chapter Security Access Service.

Service	Service Name and Required Subfunction	Subfunction	Usage
$27	**SecurityAccess**		**optional**
	requestSeed	0x01, 0x03, 0x05,0x07	**optional**
	sendKey	0x02, 0x04, 0x06, 0x08	**optional**

Table 4 : Using of Service $27

3.2 Basic Session of Security Access

The Security Access starts with a Request from the Tester (Client) to the ECU (Server).

- After Tester Request received, ECU shall generate a Random Number (RND). The RND generated is the Seed.
- The ECU will send generated Seed (RND) to Tester.
- Now, ECU and Tester has the same SEED and using special Crypto Routine ECU & Tester calculate the Key.
- The Tester send its calculated Key to the ECU.
- If the Key calculated by Tester matches the Key calculated by the ECU, then the ECU shall be unlocked for additional tasks.

This process is also known as Authorization. For furthermore information please refer also the ISO 14229 Service $27.

3.3 The Security Access

In Customer authorization process there are 4 special Levels valid. Please refer to [QD1]

Security Level	Description	Key Must Defined By	Usage
0	No Level needed for OBD relevant amounts	No Key needed	M
1	After Sales	RND & OEM Key	M
2	Manufacturing	RND & OEM Key	M
3	ECU- Coding	RND & OEM Key	O
4	ECU-Programming (Flash)	RND & OEM Key	O

Table 5 : Security Level Table 1

Security level	Description	Request Seed	Send Key	Session	Usage
0	No Level needed for OBD relevant amounts			Default	M
1	After Sales	0x01	0x02	Extended	M
2	Manufacturing	0x03	0x04	Extended	M
3	ECU- Coding	0x05	0x06	Coding	O
4	ECU-Programming (Flash)	0x07	0x08	Programming	O

Table 6 : Security Level Table 2

For the calculation in the authorization process, 4 additional Keys (one 8 Byte OEM Key per level) included. This means, for any Levels, a separate, very special Key will be calculated. The normal Security Access process is described in the picture below.

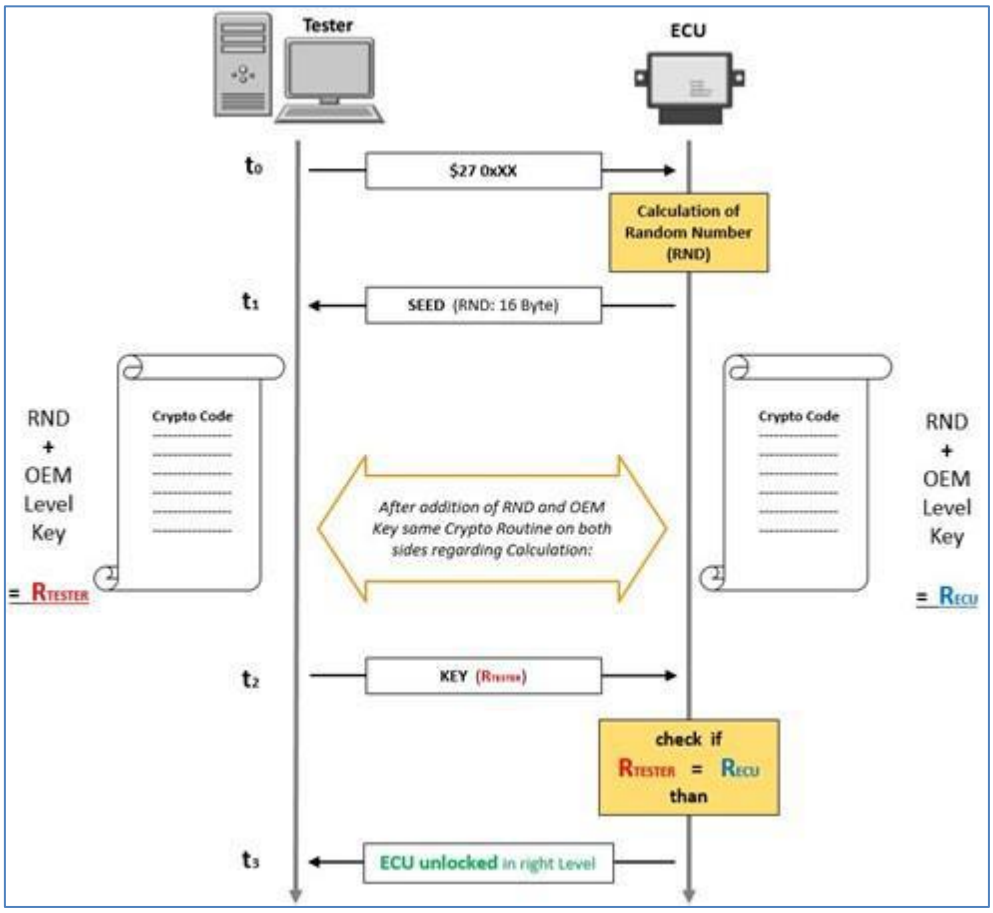


Figure 1 : Security Access Session

3.4 Step – 1: Request Service for security Access

The Tester shall send a Request Service $27 with the required Subservice (Level depending) to the ECU, start with timepoint t_0. The Subservices are described below:

After Sales	Seed 0x01	Level-1
Manufacturing	Seed 0x03	Level-2
ECU- Coding	Seed 0x05	Level-3
ECU-Programming (Flash)	Seed 0x07	Level-4

Table 7 : Request Seed & Level

3.5 Step – 2: Random Number

The ECU shall calculate a Random Number, after having received the request at timepoint "t_0" also called as RND. Therefore, the ECU needs a Random Number Generator. The function of this Generator is not the content of this Specification. This RND is the SEED and the ECU shall send it back to Tester at time point "t_1".

Note: for the project, the RND length is required with 16 Byte.

Time Period from t_0 till t_1 shall not exceed 3000ms.

3.6 Step 3 Calculation of the Key in Tester and ECU

The 8 Bytes OEM Keys are defined by OEM. For every Level a separated OEM Key is necessary. These OEM Keys shall be known on Tester side and ECU side and shall be used for the calculation.

Example for the structure of an OEM Key:

*"OEMAuto X = OEM **Level X**" KeyArray[8] =*
{
0xXX,0xXX,0xXX,0xXX,0xXX,0xXX,0xXX,0xXX
}

3.7 Step 4 Tester sends back calculated key

The Tester shall send back the calculated Key with the required Subservice "Security Access Send Key" (Level depending) to the ECU at time point t_2. For the different Levels shall be used in the Subservices as below:

After Sales	Key 0x02	Level-1
Manufacturing	Key 0x04	Level-2
ECU- Coding	Key 0x06	Level-3
ECU-Programming (Flash)	Key 0x08	Level-4

Table 8 : Response Key & Level

Only if the key is calculated identically and correctly on both sides, the ECU shall be unlocked at timepoint t_3. The ECU shall confirm this state by sending a positive response.

3.8 Security Access Mechanism

The length of Seed and Key both shall be **128bits (16 Bytes)**. The **HMAC-SHA1** Algorithm is used to calculate the key.

1. As the response to **SecurityAccess.RequestSeed,** the ECU shall generate random number (RND) and shall send generated random number (RND) to the tester. The length of this random number (RND) i.e. Seed shall be **128bits (16 bytes).**
2. The tester and ECU both have the same Seed (random number) after above step 1 and both shall calculate the HMAC value (Key) using this Seed.
3. The key calculated by tester is transmitted to the ECU by using **SecurityAccess.SendKey**.
4. If the key is calculated by tester matches to the key calculated by ECU, then the ECU shall be unlocked until it returns to diagnostic default session. The ECU shall confirm this state by sending a positive response.

5. If some mistakes occur in this process or key doesn't matches, ECU shall send a negative response with a specific NRC (Negative response code) value.

3.9 MAC-SHA1 Algorithm

Once the Seed is received, it shall start a binary addition process of both the RND and the OEM Key in the Tester and in the ECU (timepoint t_1).

There is some key word to explain in the HMAC algorithm.

H	**=**	It means a kind of hash algorithm, we use SHA1 algorithm.
B	**=**	It is the byte-length of blocks (**B** = 64 at here).
K	**=**	It is 64 bits input of the HMAC algorithm. stored in ROM.
ipad	**=**	the byte 0x36 repeated B times
opad	**=**	the byte 0x5C repeated B times

To calculate HMAC over the '**Seed**' we perform;

$$\boldsymbol{Key} = \boldsymbol{H}(\boldsymbol{K}\ XOR\ \boldsymbol{opad}, \boldsymbol{H}(\boldsymbol{K}\ XOR\ \boldsymbol{ipad}, \boldsymbol{Seed}))$$

Namely,

(1) append zeros to the end of **K** to create a **B** byte string (e.g., if **K** is of length 20 bytes and **B**=64, then **K** will be appended with 44 zero bytes 0x00)
(2) XOR (bitwise exclusive-OR) the **B** byte string computed in step with **ipad**
(3) append the stream of data 'Seed' to the **B** byte string resulting from step (2)
(4) apply **H** to the stream generated in step (3)
(5) XOR (bitwise exclusive-OR) the B byte string computed in step (1) with opad
(6) append the **H** result from step (4) to the **B** byte string resulting from step (5)
(7) apply **H** to the stream generated in step (6) and output the result
(8) truncate the 160 bits result. Output the leftmost 128 bits of the HMAC, it is the 'Key'

3.10 HA-1 Calculation

About Message Pad:
SHA-1 is used to compute a message digest for a message or data file that is provided as input. The message or data file should be considered to be a bit string. The length of the message is the number of bits in the message (the empty message has length 0).

Suppose a message has length L< 2^64. Before it is input to theSHA-1, the message is padded on the right as follows:

a) "1" is appended. Example: if the original message is "0101 0000", this is padded to "0101 0000 1".
b) "0"s are appended. The number of "0"s will depend on the original length of the message. The last 64 bits of the last 512-bit block are reserved for the length L of the original message.

c) Obtain the 2-word representation of L, the number of bits in the original message. If L < 2^32 then the first word is all zeroes. Append these two words to the padded message. (note that L is computed before any padding)

The padded message will contain 16 * n words for some n > 0. The padded message is regarded as a sequence of n blocks M(1) , M(2), first characters (or bits) of the message.

Some Function and Constants Used:

f(t;B,C,D) is defined as follows: for words(32-bit) B, C, D,

f(t;B,C,D) = (B AND C) OR ((NOT B) AND D) (0 <= t <= 19)
f(t;B,C,D) = B XOR C XOR D (20 <= t <= 39)
f(t;B,C,D) = (B AND C) OR (B AND D) OR (C AND D) (40 <= t <= 59)
f(t;B,C,D) = B XOR C XOR D (60 <= t <= 79).

A sequence of constant words K (0), K (1), ..., K(79) is used in the SHA-1. In hex these are given by

K(t) = 0x5A827999 (0 <= t <= 19)
K(t) = 0x 6ED9EBA1 (20 <= t <= 39)
K(t) = 0x 8F1BBCDC (40 <= t <= 59)
K(t) = 0x CA62C1D6 (60 <= t <= 79).

Let MASK = 0x0000000F.

H0 = 0x67452301
H1 = 0xEFCDAB89
H2 = 0x98BADCFE
H3 = 0x10325476
H4 = 0xC3D2E1F0.

Then processing of M(i) is as follows:

a) Divide M(i) into 16 words W[0], ... , W[15], where W[0] is the left-most word
b) Let A = H0, B = H1, C = H2, D = H3, E = H4
c) For t = 0 to 79 do s = t AND MASK;
   if (t >= 16)
   W[s] = S^1(W [(s + 13) AND MASK] XOR W [(s + 8) AND MASK] XOR W[(s + 2) AND MASK] XOR W[s]);
   TEMP = S^5(A) + f(t;B,C,D) + E + W[s] + K(t); E = D; D = C; C = S^30(B); B = A; A = TEMP;
d) Let H0 = H0 + A, H1 = H1 + B, H2 = H2 + C, H3 = H3 + D, H4 = H4 + E.

After processing M(n), the message digest is the 160-bit string represented by the 5 words.
H0 H1 H2 H3 H4.

3.11 Mechanism of the CRC 32 Algorithm

To, verify correct flash programming a CRC32 value shall be calculated by the follow generator polynomial:

$$x32 + x26 + x23 + x22 + x16 + x12 + x11 + x10 + x8 + x7 + x5 + x4 + x2 + x + 1$$

The initial value of the shift register shall be 0xFFFFFFFF and the final register value will be logically inverted before being returned.

For performance reasons, a table look-up method should be used for simplifying the algorithm complexity.

3.12 Failure Behavior

4 Standards

4.1 Referenced Documents

Ref.No.	Name	Version
[QD1]	CES_VFDSP50000033_EEP_Basic_Diagnostics_Specification_	latest

Table 9: Referenced Documents

4.2 Norms and Standards

Ref.No.	Name
SAE J1850	Class B Data Communications Network Interface
RFC 3174	US Secure Hash Algorithm1 (SHA1)
RFC 2104	HMAC: Keyed-Hashing for Message Authentication
ANSI x9.71-2000	Keyed Hash Message Authentication Code

Table 10 : Normative references

This Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

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